

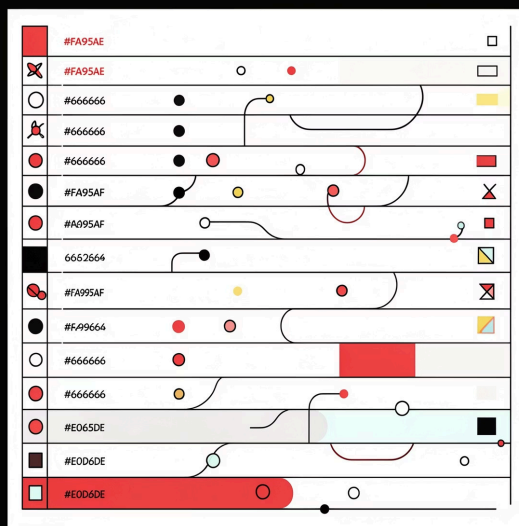
Customer Behaviour Analysis

A deep dive into retail e-commerce data — uncovering purchase patterns, customer segments, and revenue drivers to support smarter business decisions.

RETAIL · E-COMMERCE · DATA ANALYTICS



About the Dataset



Dataset Overview

- **Type:** Retail / E-commerce — Customer Transaction Level
- **Numerical:** Age, Purchase Amount, Previous Purchases
- **Categorical:** Category, Payment Method, Season

i Used for **customer behavior analysis** and **business decision-making**

Why This Data Matters

Identify High-Value Customers

Pinpoint who drives the most revenue

Optimize Discounts

Understand which promotions actually work

Improve Retention

Increase customer lifetime value (CLV)

Boost Campaigns

Target the right customers at the right time



Business Problem

How can a retail company understand customer purchasing behavior to **increase revenue**, improve **customer retention**, and optimize **marketing strategies**?

Low Retention

Who are frequent vs. one-time buyers?

Ineffective Campaigns

Do discounts and promo codes actually drive purchases?

Demand & Segmentation

Which categories, seasons, and customer types generate the most value?

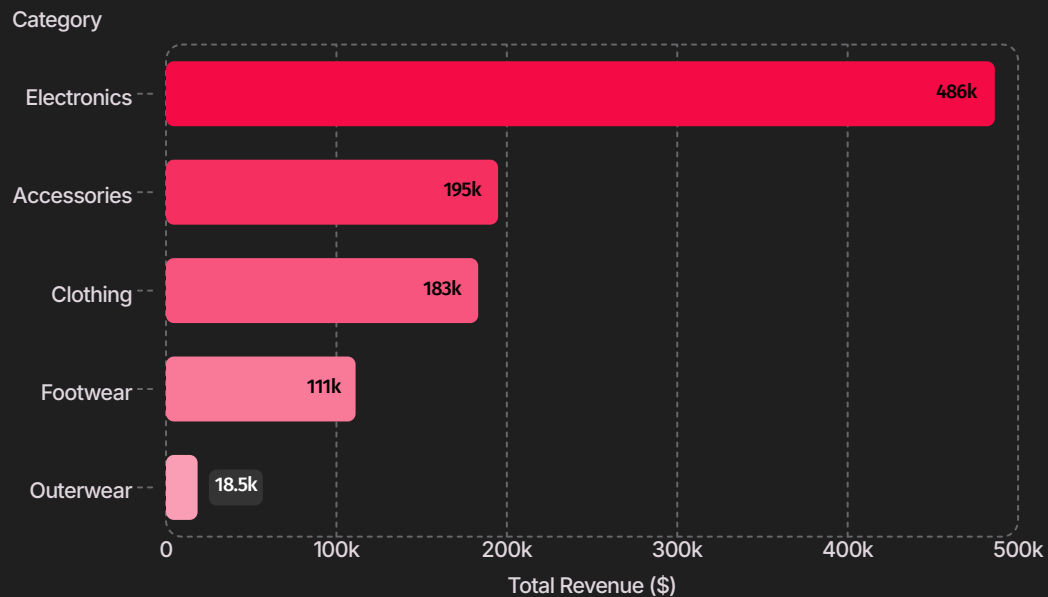
Exploratory Data Analysis (EDA)



Key Cleaning Steps

- **Category correction:** Mapped items to correct categories
- **Missing values:** Domain-logic fills — e.g., Electronics → "Not Applicable," Clothing → mode
- **Review Rating:** Filled using product-level mean (advanced imputation)
- **Previous Purchases:** Nulls replaced with 0 (no prior purchase)
- **Duplicates:** Removed by Customer ID
- **Column names:** Standardized to snake_case for SQL & Power BI

SQL Analysis: Revenue Insights



Key Findings

- **Electronics** dominates with \$486K — nearly 2.5× Accessories
- Discounted customers average **\$219 spend** vs. \$182 without discount
- **Male** customers lead revenue at \$391K; Female \$352K; Other \$251K
- **51+ age group** contributes 40.2% of total revenue (\$399K)

SQL Analysis: Customer Segments & Subscriptions

3,085

Loyal Customers

Most valuable segment

1,354

Returning Customers

561

New Customers

Subscription Impact

Status	Customers	Avg Spend
Subscribed	1,579	\$270.58
Not Subscribed	3,421	\$165.72

Subscribers spend **63% more** on average. Repeat buyers (5+ purchases) are also more likely to subscribe.

Top & Bottom Performers

★ Top 5 by Rating

Item	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.79

📉 Bottom 5 by Rating

Item	Avg Rating
Phone	2.94
Headphones	3.08
Watch	3.10

💰 Top 5 by Revenue

Item	Revenue
Headphones	\$138K
Phone	\$126K
Bag	\$121K
Laptop	\$115K
Watch	\$108K

⚠️ Electronics generate high revenue but score **lowest in ratings** — a key quality gap to address.

Power BI Dashboard



5,000

Total Customers

\$994.18K

Total Spend

\$198.84

Average Spend

3.62

Average Rating

Key Takeaways

→ Electronics = Revenue King

Highest revenue category (\$486K) but lowest customer satisfaction — quality improvement is critical.

→ Discounts Drive Higher Spend

Discounted customers spend \$219 avg vs. \$182 — targeted promotions are worth investing in.

→ Subscriptions Are Powerful

Subscribers spend 63% more. Growing the subscriber base is a high-ROI retention strategy.

→ 51+ Customers Dominate Revenue

Older customers contribute 40% of total spend — a segment worth prioritizing in campaigns.

